Step 1 — Import Libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
from sklearn.model_selection import train_test_split
```

Step 2 — Load Dataset

```
from google.colab import drive
drive.mount('/content/gdrive')
```

```
path='/content/gdrive/MyDrive/Knowledge_Score_data.csv'
df = pd.read_csv(path)
```

Preview dataset

```
df.head()
```

```
df.isnull().sum()
```

Step 3 — Check Correlation Between Features

```
sns.pairplot(df)
plt.show()
```

Step 4 — Select Features and Target

```
X = df.drop("Knowledge_Score", axis = 1)
y = df["Knowledge_Score"]
```

Step 5 — Train Test Split

Training dataset 80% , test dataset 20%

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

Step 6 — Model Selection and Training

```
model = LinearRegression()

model.fit(X_train, y_train)
```

Step 7 — Model Prediction

```
y_pred = model.predict(X_test)
```

Step 8 — Model Evaluation

```
#MSE
mse = mean_squared_error(y_test, y_pred)
print("MSE:", mse)
```

```
#RMSE
rmse = np.sqrt(mse)
print("RMSE:", rmse)
```

```
#R2-score
r2 = r2_score(y_test, y_pred)
print("R2 Score:", r2)
```

```
#Actual Vs Predicted
plt.scatter(y_test, y_pred)
plt.xlabel("Actual Knowledge Score")
plt.ylabel("Predicted Knowledge Score")
plt.title("Actual vs Predicted")
plt.show()
```

Step 9 — Model Coefficients

```
coefficients = pd.DataFrame({
    "Feature": X.columns,
    "Coefficient": model.coef_
})

print(coefficients)
```

```
intercept = model.intercept_
print(intercept)
```

SNS Regression Plot

```
sns.regplot(data = df, x = "Number of pages", y = "Knowledge_Score", color = "red")
sns.scatterplot(data = df, x = "Number of pages", y = "Knowledge_Score")
plt.xlabel("Diet Quality")
plt.ylabel("Health Score")
plt.title("Correlation between Number of pages and Knowledge Score")
plt.show()
```

Applying Regularizations

```
from sklearn.linear_model import Ridge, Lasso

ridge = Ridge()

ridge.fit(X_train, y_train)
y_pred_Rd = ridge.predict(X_test)

plt.scatter(y_test, y_pred_Rd )
plt.xlabel("Values")
plt.ylabel("Predictions")
plt.title("Correlation between Values and Predictions")
plt.show()


print(f"MSE\t: {mean_squared_error(y_test, y_pred_Rd )}")
print(f"R2 Score: {r2_score(y_test, y_pred_Rd )}")
```

```
lasso = Lasso()
```

```
lasso.fit(X_train, y_train)
y_pred_LS = lasso.predict(X_test)

plt.scatter(y_test, y_pred_LS )
plt.xlabel("Values")
plt.ylabel("Predictions")
plt.title("Correlation between Values and Predictions")
plt.show()


print(f"MSE\t: {mean_squared_error(y_test, y_pred_LS )}")
print(f"R2 Score: {r2_score(y_test, y_pred_LS )}")
```

```
#Check if the model overfits
print("Train score:", model.score(X_train, y_train))
print("Test score:", model.score(X_test, y_test))
```